

Workflow Manual & QA Guide

Every workflow the system runs, written as a scenario with its data flow across roles — plus a concrete QA walkthrough and expected result for each, so the testing team has ready-made test cases.

Roles: Admin · Custodian · Maintenance Personnel (MP) **Focus:** Vehicle availability **Version:** 2.0

● **Admin**

Manages vehicles & users, creates and confirms tickets, assigns mechanics, and oversees all records.

● **Custodian**

Reports issues, runs condition checks, inspects vehicles, and verifies completed repairs.

● **Maintenance Personnel**

Receives work orders, performs and logs repairs, and records maintenance activities.

How to read this guide: each workflow has a **Scenario** (a concrete QA walkthrough with the expected result), a **Data flow** ([ROLE] = who acts, ---> = next step, **bold** = resulting status, *branch* = a decision), and a **Note** clarifying how the real system behaves. QA cast used below: **Roel** (Admin), **Nicole** (Custodian), **Toto** (Maintenance).

01 Staff Reports a Vehicle Defect

CUSTODIAN

ADMIN

MP

Someone files a defect; the Admin reviews it and, if real, opens a maintenance ticket that runs all the way through repair and closure. This is the system's main workflow.

SCENARIO — QA WALKTHROUGH

Nicole notices Ambulance **AMB-101**'s brakes feel weak. She files an Issue Report (Brake Issue, **High**, with a photo). Roel reviews it and clicks **Create Ticket**, assigning Nicole to inspect. Nicole inspects → **Needs Maintenance**. Roel assigns Toto + work order. Toto replaces the pads and logs the repair. Nicole verifies → **Approved**. Roel confirms → **Confirmed**.

Expected: ticket becomes **Done** and moves to Ticket Archives; AMB-101 returns to **Available / Good**; the linked issue is marked **Resolved**; a maintenance record is auto-created.

DATA FLOW

```
[ CUSTODIAN ] ---> submits an Issue Report (type, severity, photo) ---> vehicle →  
Needs Inspection ---> [ ADMIN ] ---> reviews; Reject = leave/delete, Approve =  
"Create Ticket" ---> assigns ticket to a [ CUSTODIAN ] for inspection ---> inspects  
(Needs Maintenance / No Issues) ---> [ ADMIN ] ---> assigns [ MP ] + work order ---  
> fixes & logs ---> [ CUSTODIAN ] ---> verifies (Approve / Reject → back to MP) ---  
> [ ADMIN ] ---> confirms (Approve = closed & archived, vehicle → Available /  
Reject = reopen)
```

Note: the system inserts a **Custodian inspection step before the mechanic is assigned**; the front "validate" isn't a formal gate — creating the ticket is the approval. A "No Issues" inspection skips the repair and goes straight to the Admin's confirmation.

02 Custodian Condition Check

CUSTODIAN

ADMIN

A routine check of a vehicle's physical state so problems are caught early and availability stays accurate.

SCENARIO — QA WALKTHROUGH

During the morning check, Nicole records Fire Truck **FIRE-01** with condition result **Needs Repair** (coolant leak) and adds an observation.

Expected: FIRE-01 immediately shows **Under Maintenance** and drops off the "available now" count; **no ticket is created automatically**; the check appears in the vehicle's history.

DATA FLOW

```
[ CUSTODIAN ] ---> logs a condition check (Good / Needs Inspection / Needs Repair / Damaged) ---> [ SYSTEM ] ---> IF Needs Repair or Damaged → vehicle → Under Maintenance; ELSE stays Available
```

Note: a condition check is a **standalone log** — it does not open a ticket by itself. If a repair is needed, it is escalated by filing an Issue Report (Workflow 01).

03 Repair Fails Verification (Rework Loop)

MP

CUSTODIAN

ADMIN

A mechanic finishes a repair, but the custodian's check shows it isn't actually fixed, so it goes back for rework instead of returning to service.

SCENARIO — QA WALKTHROUGH

Toto marks AMB-101's brake repair done. Nicole test-checks it — still weak — and clicks **Reject** with a note. Toto finds a stuck caliper, fixes it, and re-logs. Nicole re-verifies → **Approved**; Roel confirms.

Expected: on Reject the ticket returns to **Under Repair** and Toto is notified; the vehicle stays Under Maintenance throughout; only after Approve + Confirm does it become **Done** and Available.

DATA FLOW

```
[ MP ] ---> logs repair done ---> ticket → For Inspection ---> [ CUSTODIAN ] --->
Reject ---> ticket → Under Repair ---> [ MP ] ---> re-does & re-logs --->
[ CUSTODIAN ] ---> re-verifies (loops until Approved) ---> [ ADMIN ] ---> confirms
---> Done
```

Note: the same reject-and-loop exists at the Admin's confirmation step — a **Reopen** sends the ticket back to inspection or repair.

04 Admin Cancels / Restores a Ticket ADMIN

A ticket was opened by mistake, or the problem turned out to be nothing. The Admin cancels it and the vehicle immediately returns to service.

SCENARIO — QA WALKTHROUGH

Roel opens a ticket for Patrol multicab **PAT-202** thinking the engine light means trouble — it's a loose gas cap. He clicks **Cancel**. Realizing he cancelled the wrong ticket, he clicks **Un-cancel**.

Expected: after Cancel, PAT-202 is **Available** and the linked issue reverts to **Pending**; after Un-cancel, the ticket is restored to the **exact stage it was in** before cancelling.

DATA FLOW

```
[ ADMIN ] ---> Cancel (any active stage) ---> ticket → Cancelled, vehicle → Available, linked issue → Pending ---> IF cancelled by mistake ---> [ ADMIN ] ---> Un-cancel ---> ticket restored to its exact previous stage
```

Note: Cancel is available at every stage except Done/Cancelled. The Admin can also **Delete** a ticket entirely for genuine mistakes.

05 Direct Maintenance Record

MP

CUSTODIAN

ADMIN

A minor, external, or historical repair is logged directly — without the full ticket workflow — so the vehicle's maintenance history stays complete.

SCENARIO — QA WALKTHROUGH

Toto logs a direct oil change on Dump truck **DMP-303** as a maintenance record, then sets its progress to **Completed**. Nicole verifies → **Passed**; Roel confirms.

Expected: when Toto marks "Completed" it does **not** close — it becomes **For Verification** (a mechanic can't self-close). Only after Nicole passes it and Roel confirms does it become **Completed** and DMP-303 return to **Available**.

DATA FLOW

```
[ MP / ADMIN ] ---> adds a maintenance record ---> vehicle → Under Maintenance --->
[ MP ] ---> marks Completed ---> auto → For Verification ---> [ CUSTODIAN ] --->
verifies (Passed / Failed → Under Repair) ---> [ ADMIN ] ---> confirms --->
Completed, vehicle → Available
```

Note: a mechanic **cannot close their own work** — marking "Completed" always drops to For Verification. Every ticket confirmed Done is also written here automatically, so this is the single complete maintenance history.

06 Preventive Maintenance Scheduling

ADMIN

MP

To prevent breakdowns, routine servicing is planned ahead on the calendar so it isn't forgotten.

SCENARIO — QA WALKTHROUGH

Roel books an oil change for **AMB-101** with a **future** date, and another with a **past** date (to test overdue).

Expected: the future one is counted under **Upcoming Maintenance** on the dashboard; the past-due one is counted under **Overdue Maintenance** and highlighted in the Availability Forecast panel.

DATA FLOW

[ADMIN / MP] ---> books a schedule (vehicle, type, date) ---> status **Scheduled** --
-> [SYSTEM] ---> shows on Dashboard as **Upcoming** (due soon) or **Overdue** (past due)
---> [ADMIN / MP] ---> marks **Completed** / **Cancelled**

Note: a schedule is a **plan only** — it does not auto-create a ticket. The maintenance is started manually when it comes due.

07 Availability Forecast NEW MP ADMIN

Beyond "what's available now," the Admin needs to know **when** vehicles under repair will be back, to plan ahead.

SCENARIO — QA WALKTHROUGH

AMB-101 is Under Maintenance. While logging repairs, Toto sets **Estimated Return Date = Jul 14**. Roel opens the Dashboard, then later confirms the finished repair.

Expected: the **Availability Forecast** panel lists "AMB-101 — Ready by Jul 14, 2026" and the "ready now" count excludes it. After Roel confirms the repair, AMB-101 returns to **Available** and disappears from the forecast (the estimate auto-clears).

DATA FLOW

```
[ MP ] ---> sets an Estimated Return Date when logging repairs (or [ ADMIN ] via the vehicle's Edit form) ---> [ SYSTEM ] ---> projects an available-on date ---> [ ADMIN ] ---> reads the Dashboard forecast (X ready now + each vehicle "ready by <date>") ---> vehicle returns to service ---> estimate auto-clears
```

Note: only vehicles currently Under Maintenance appear in the forecast, and the estimate clears automatically the moment a vehicle is confirmed back in service — so the forecast is never stale.

08 Register a Vehicle ADMIN

The barangay acquires a vehicle; the Admin registers it so it joins the fleet and can be monitored.

SCENARIO — QA WALKTHROUGH

Roel adds “Ambulance 9” (plate **AFAD123**) via the 3-step form: Basic Info → Specs (with capacity unit and fuel type) → Photo & Location.

Expected: the vehicle appears in Vehicle Management as **Available / Good**, with its uploaded photo; a duplicate plate number or an invalid plate format is rejected with a clear message.

DATA FLOW

[**ADMIN**] ---> Add Vehicle (Step 1 Basic Info → Step 2 Specs → Step 3 Photo & Location) ---> [**SYSTEM**] ---> vehicle saved as **Available / Good**

Note: the photo is stored in cloud storage; a newly registered vehicle always starts as Available.

09 Retire / Restore a Vehicle ADMIN

An end-of-life vehicle is removed from the available fleet, but its records are preserved — and it can be brought back if needed.

SCENARIO — QA WALKTHROUGH

Roel archives the 18-year-old dump truck **DMP-303**, then later restores it.

Expected: after Archive, DMP-303 becomes **Inactive**, is removed from available/total-active counts, but its history and past records remain viewable; after Restore, it returns to **Available**.

DATA FLOW

[**ADMIN**] ---> Archive ---> vehicle → **Inactive** (off the available list, full history kept) ---> IF needed again ---> [**ADMIN**] ---> Restore ---> vehicle → **Available**

10 Multiple Defects on One Vehicle

CUSTODIAN

ADMIN

The system keeps **one active maintenance ticket per vehicle**, so a second defect can't spawn a competing ticket while one is already in progress.

SCENARIO — QA WALKTHROUGH

Nicole reports a brake issue on **AMB-101** and Roel opens a ticket. Two days later another defect (broken aircon) is reported on the same AMB-101, and Roel tries to create a second ticket.

Expected: the system **blocks the second active ticket** with a clear message ("this vehicle already has an active maintenance ticket"); the aircon issue stays **Pending** until the first ticket closes, or its details are added to the open ticket.

DATA FLOW

```
[ CUSTODIAN ] ---> files defect #1 ---> [ ADMIN ] ---> creates ticket --->
[ ANYONE ] ---> files defect #2 on the same vehicle ---> [ SYSTEM ] ---> blocks a
2nd active ticket (one active ticket per vehicle)
```

Note: this is **by design** — it prevents two conflicting repairs on the same vehicle. The same guard applies when reopening an archived ticket.

11 Admin Oversight & Accountability ADMIN

The Admin monitors the whole fleet, prepares summaries for the barangay council, and manages who can use the system.

SCENARIO — QA WALKTHROUGH

Roel opens **Reports** and generates a maintenance summary, checks **Logs** to see recent actions, and creates a new **Custodian** account under **Users**.

Expected: reports render with the correct records; Logs show the latest actions with user, role, and time; the new user can sign in and sees only the Custodian menu; deactivating a user blocks their login.

DATA FLOW

[**ADMIN**] ---> Dashboard (fleet status, issues, upcoming/overdue maintenance, forecast) ---> Reports (vehicle / issue / maintenance / history) ---> Vehicle History (per-vehicle audit trail) ---> Logs (who did what, when) ---> Users (create accounts, activate / deactivate, assign role)

Note: every action across the system is recorded automatically in Vehicle History and Logs, and notifications are sent to the next person at each handoff.